LESSON 5.8

Determine Surface Area of Pyramids

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the surface area of a pyramid by adding the base area and the lateral faces.

WHY THIS MATTERS Your class is building a pyramid-shaped display to showcase the time capsule at the school entrance. The pyramid will be covered in gold leaf to make it shine. To figure out how much gold leaf you need, you must calculate the total surface area — the base plus all the triangular faces! The pyramid has a square base that is 10 inches on each side, and each triangular face has a slant height of 12 inches.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Pyramid *Spanish: Pirámide*	A solid with a flat bottom and triangle sides that meet at one point on top.	square base + 4 triangles
Slant height *Spanish: Altura inclinada (apotema)*	The height of a side triangle, measured along its slanted face.	used in ½ × b × slant
Lateral face *Spanish: Cara lateral*	A triangle side of a pyramid, not the bottom.	4 on a square pyramid
Base *Spanish: Base*	The flat bottom of a pyramid.	the square on the net
Apex *Spanish: Ápice*	The point at the top of a pyramid where the sides meet.	tip of the pyramid
Lateral area *Spanish: Área lateral*	The total area of just the side triangles, not the bottom.	For a square pyramid: lateral area = 4 × (½ × base edge × slant height)

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Square base 10 cm; slant height 7 cm.

1. Base: 10 × 10 = 100 cm².
2. One triangle: ½ × 10 × 7 = 35 cm².
3. Four triangles: 140. Add: 100 + 140.

Answer: SA = 240 cm²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller square pyramid: base 4 × 4, slant height 5. What is the base area?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The base is a _____ with an area of _____ .
- Each triangular face has a base of _____ and a slant height of _____ .
- There are _____ triangular faces, so together they cover _____ .
- The surface area is _____ because I added _____ .

Word bank pyramid | base | slant height | lateral face | apex | triangle | square | congruent | add | surface area | square units | net

WATCH OUT!

A common mistake

A common mistake in Surface Area of Pyramids is finding the four triangular lateral faces correctly (½ × base edge × slant height, times 4) and then stopping

LESSON 5.8

Determine Surface Area of Pyramids

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the area of one lateral face for each pyramid.

Pyramid	Base Edge	Slant Height	Area of One Face
Pyramid 1	4 in	9 in	
Pyramid 2	6 cm	5 cm	
Pyramid 3	10 ft	8 ft	

HINT Each row asks for the area of ONE triangular face, using its base edge and slant height.

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Put the steps in order for finding the surface area of a square pyramid (base edge 6, slant height 10).

Groups:

_____ Find base area: 6 × 6 = 36

_____ Find one lateral face: ½ × 6 × 10 = 30

_____ Multiply by 4 faces: 4 × 30 = 120

_____ Add base + lateral: 36 + 120

_____ SA = 156 in²

HINT The cards walk from base area to lateral faces to the final total for a base-6, slant-10 pyramid.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A square pyramid has a base edge of 5 in and a slant height of 7 in. What is the surface area?

(A) 70 in²
(B) 95 in²
(C) 95 in³
(D) 120 in²

HINT A square pyramid has TWO kinds of faces: one square base (5 × 5) and four identical triangles with slant height 7.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

How many lateral (triangular) faces does a square pyramid have?

(A) 3
(B) 4
(C) 5
(D) 6

HINT Lateral faces are the slanted triangles on the sides — the square base itself doesn't count.

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

What is the area of one triangular face with base 8 in and slant height 6 in?

(A) 14 in²
(B) 24 in²
(C) 24 in³
(D) 48 in²

HINT You have one triangle: base 8 in, slant height 6 in. Triangle area is not just base × height.

Formula	_____
Put the numbers in	_____
Work it out	_____
Answer with its unit	_____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Don't Forget the Base

1. **Box** Square pyramid: base edge = 5 in, slant height = 6 in
2. **One Triangular Face** ½ × 5 × 6 = 15 in²
3. **Four Faces** 4 × 15 = 60 in²
4. **Answer** SA = 60 in² (stopped after the four triangular faces)

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Did the student include the bottom of the pyramid?

CORRECT WORK

_____

_____

_____

_____

Correct answer: _____

EXPLAIN YOUR THINKING

Display B has the largest surface area (280 in²). What contributes more to the total — the base or the lateral faces? Why?

SENTENCE STARTER For Display B, the base area is ___ in² and the total lateral area is 4 × ___ = ___ in². The ___ contribute(s) more because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.8

Determine Surface Area of Pyramids

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the surface area of each square pyramid.

Pyramid	Base Edge	Slant Height	Base Area	Total SA
Pyramid X	12 in	10 in		
Pyramid Y	8 cm	6 cm		
Pyramid Z	14 ft	11 ft		

SHOW YOUR WORK

2 **Use the model to solve. Show your work.**



Total lateral area = ? in² (SA − base = 260 − 100).

SHOW YOUR WORK

Answer: _____

3 **Write the name of the correct group on each line.**

Sort these pyramids from LEAST surface area to GREATEST.

Groups:

_____ Base 4×4, slant 5

_____ Base 6×6, slant 5

_____ Base 6×6, slant 8

_____ Base 10×10, slant 9

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A square pyramid has a base area of 49 cm² and a total lateral area of 84 cm². What is the surface area?

(A) 49 cm²

(B) 84 cm²

(C) 133 cm²

(D) 133 cm³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Did You Halve the Triangle?

1. **Box** Square pyramid: base edge = 8 in, slant height = 10 in
2. **Base Area** 8 × 8 = 64 in²
3. **One Triangular Face** 8 × 10 = 80 in² (forgot the ½)
4. **Answer** SA = 64 + 4 × 80 = 64 + 320 = 384 in²

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Display B has the largest surface area (280 in²). What contributes more to the total — the base or the lateral faces? Why?

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LESSON 5.8

Determine Surface Area of Pyramids

Challenge · Extension

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- ☐ Exceeds
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Compare a square pyramid and a rectangular prism that sit on the same 6 × 6 base.

Shape	Base Area	Lateral/Side Area	Total SA
Square pyramid (slant h = 8)	36		
Rectangular prism (h = 8)	36 (× 2 bases)		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Surface Area Error

1. **Dimensions** Square pyramid: base edge = 10 in, slant height = 8 in
2. **Base Area** 10 × 10 = 100 in²
3. **Lateral Area** ½ × 10 × 8 = 40 in² (one face). Total lateral = 40 × 3 = 120 in²
4. **Answer** SA = 100 + 120 = 220 in²

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Two pyramids have the same base area (64 in²) but different slant heights. Pyramid A has slant height 6 in and Pyramid B has slant height 10 in. How much more surface area does Pyramid B have? Why does slant height affect surface area but NOT base area?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Display B has the largest surface area (280 in²). What contributes more to the total — the base or the lateral faces? Why?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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